

MARMARA UNIVERSITY
FACULTY OF ENGINEERING



EE7027

INTRODUCTION TO MACHINE LEARNING

HOMEWORK V

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Submitted to: Prof. Cabir Vural

SOLUTION FOR QUESTION 1

a)

Calculating weights for each sample in dataset in 4 iterations can be shown as follows.

Step -1

$$w_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\widehat{y_0} = sign(w_0^T x_0) = sign\left(\begin{bmatrix} 0 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = sign(0) = -1$$

$$y_0 = 1$$

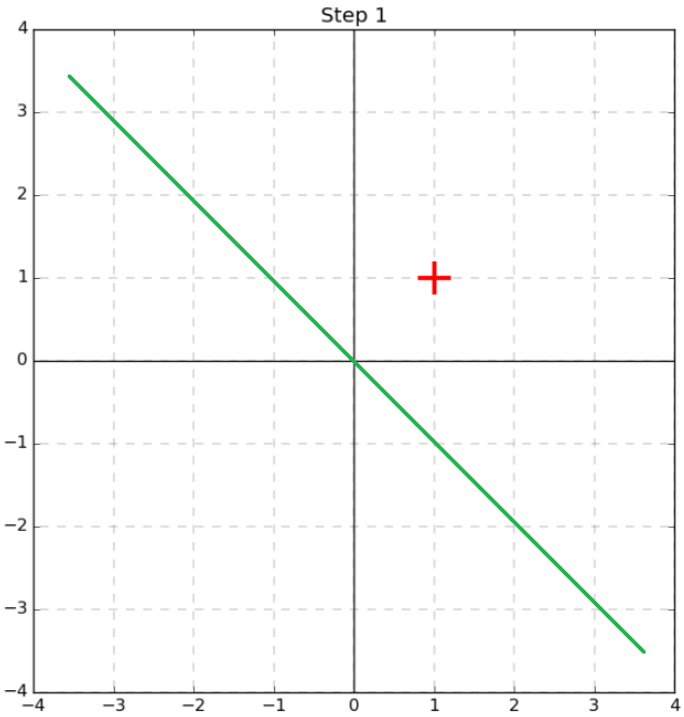
$$\widehat{y_0} \neq y_0$$

Update weights as follows

$$w_1 = w_0 + y_0 x_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The decision boundary can be written as follows

$$w^T x > 0 = \begin{bmatrix} 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 + x_2 \begin{cases} > 0, \hat{y} = 1 \\ \leq 0, \hat{y} = -1 \end{cases}$$



Step -2

$$w_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\widehat{y_1} = sign(w_1^T x_1) = sign\left(\begin{bmatrix} 1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix}\right) = sign(1) = 1$$

$$y_1 = -1$$

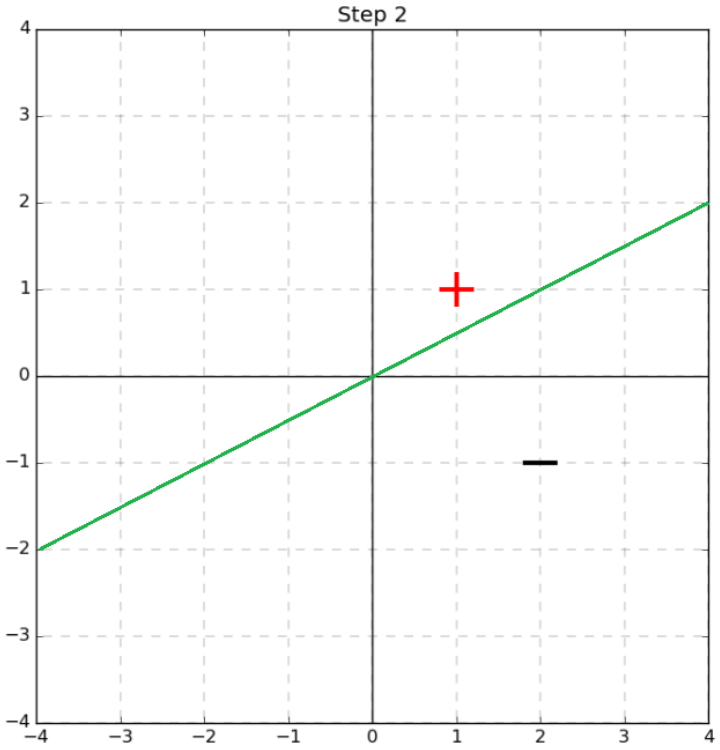
$$\widehat{y_1} \neq y_1$$

Update weights as follows

$$w_2 = w_1 + y_1 x_1 = \begin{bmatrix}1\\1\end{bmatrix} - 1\begin{bmatrix}2\\-1\end{bmatrix} = \begin{bmatrix}-1\\2\end{bmatrix}$$

The decision boundary can be written as follows

$$w^T x > 0 = [-1 \quad 2]\begin{bmatrix}x_1\\x_2\end{bmatrix} = -x_1 + 2x_2 \begin{cases} > 0, \hat{y} = 1 \\ \leq 0, \hat{y} = -1 \end{cases}$$



Step -3

$$w_2 = \begin{bmatrix}-1\\2\end{bmatrix}$$

$$\widehat{y_2} = sign(w_2^T x_2) = sign\left([-1 \quad 2]\begin{bmatrix}-3\\-1\end{bmatrix}\right) = sign(1) = 1$$

$$y_2 = -1$$

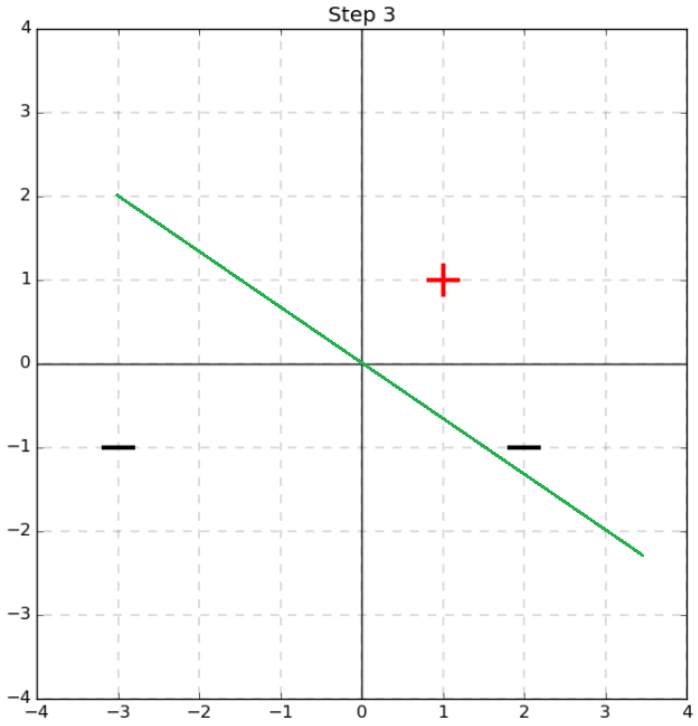
$$\widehat{y_2} \neq y_2$$

Update weights as follows

$$w_3 = w_2 + y_2 x_2 = \begin{bmatrix}-1\\2\end{bmatrix} - 1\begin{bmatrix}-3\\-1\end{bmatrix} = \begin{bmatrix}2\\3\end{bmatrix}$$

The decision boundary can be written as follows

$$w^T x > 0 = [2 \quad 3]\begin{bmatrix}x_1\\x_2\end{bmatrix} = 2x_1 + 3x_2 \begin{cases} > 0, \hat{y} = 1 \\ \leq 0, \hat{y} = -1 \end{cases}$$



Step -4

$$w_3 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\widehat{y_3} = sign(w_3^T x_3) = sign\left(\begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \end{bmatrix}\right) = sign(-3) = -1$$

$$y_3 = 1$$

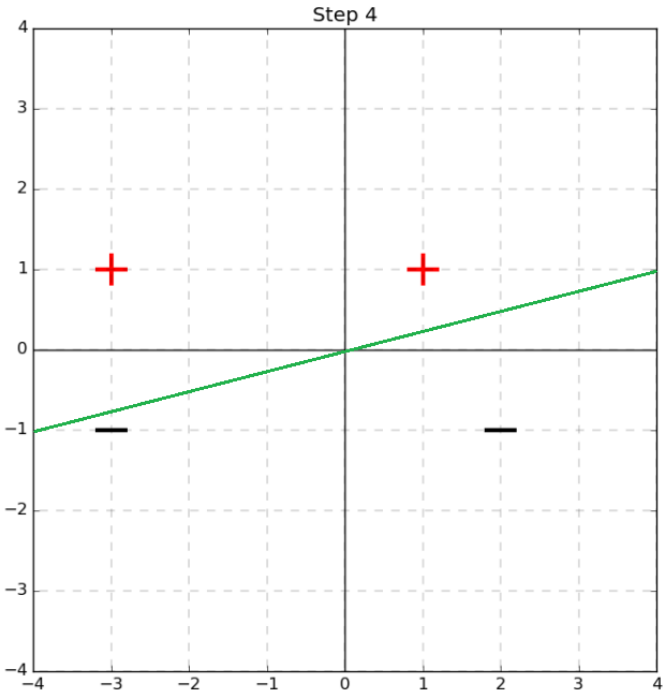
$$\widehat{y_3} \neq y_3$$

Update weights as follows

$$w_4 = w_3 + y_3 x_4 = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 1 \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

The decision boundary can be written as follows

$$w^T x > 0 = \begin{bmatrix} -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -1x_1 + 4x_2 \begin{cases} > 0, \hat{y} = 1 \\ \leq 0, \hat{y} = -1 \end{cases}$$



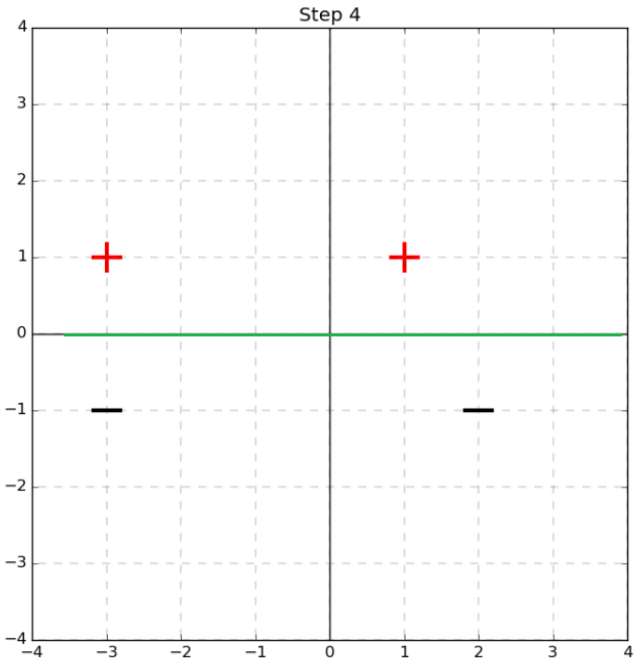
b)

This learned perceptron do not maximize the margin between the training data and the decision boundary when the boundary is $-x_1 + 4x_2 = 0$. We can see that by calculating minimum distance for each label to the boundary itself as follows

$$d_+ = \frac{|-1+4|}{\sqrt{1^2+4^2}} = \frac{3}{\sqrt{17}} \approx 0.72$$

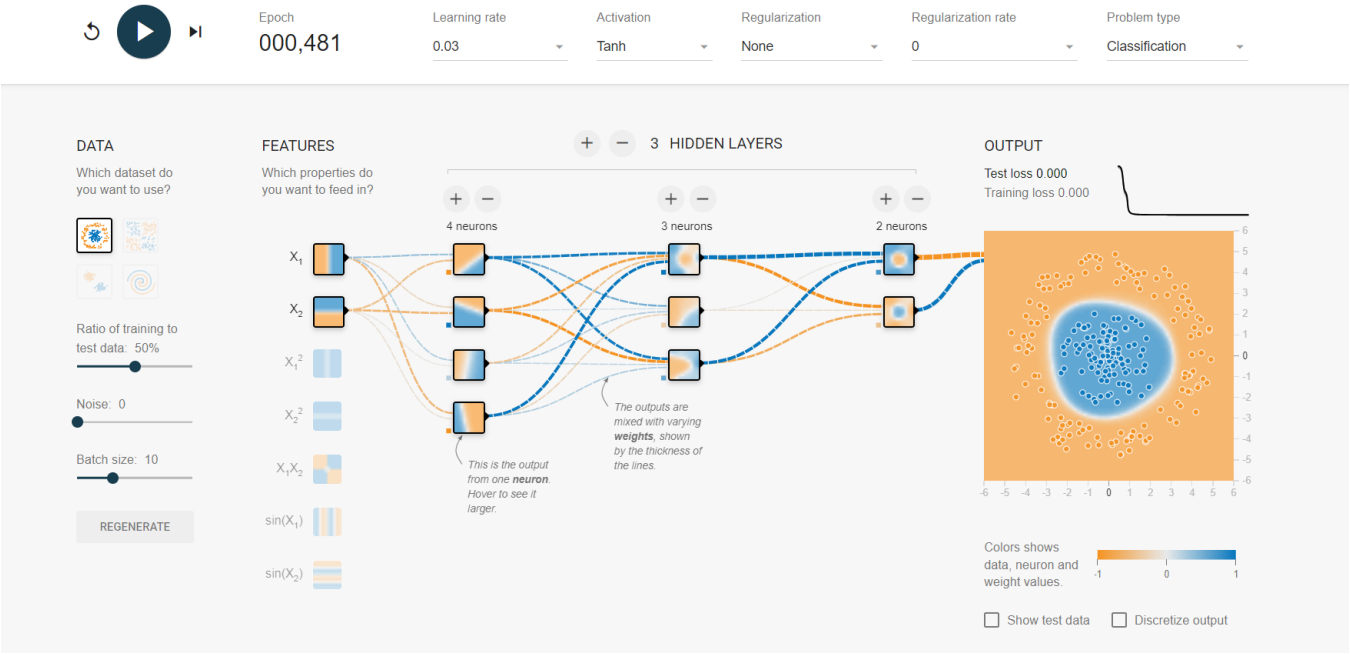
$$d_- = \frac{|3-4|}{\sqrt{1^2+4^2}} = \frac{1}{\sqrt{17}} \approx 0.24$$

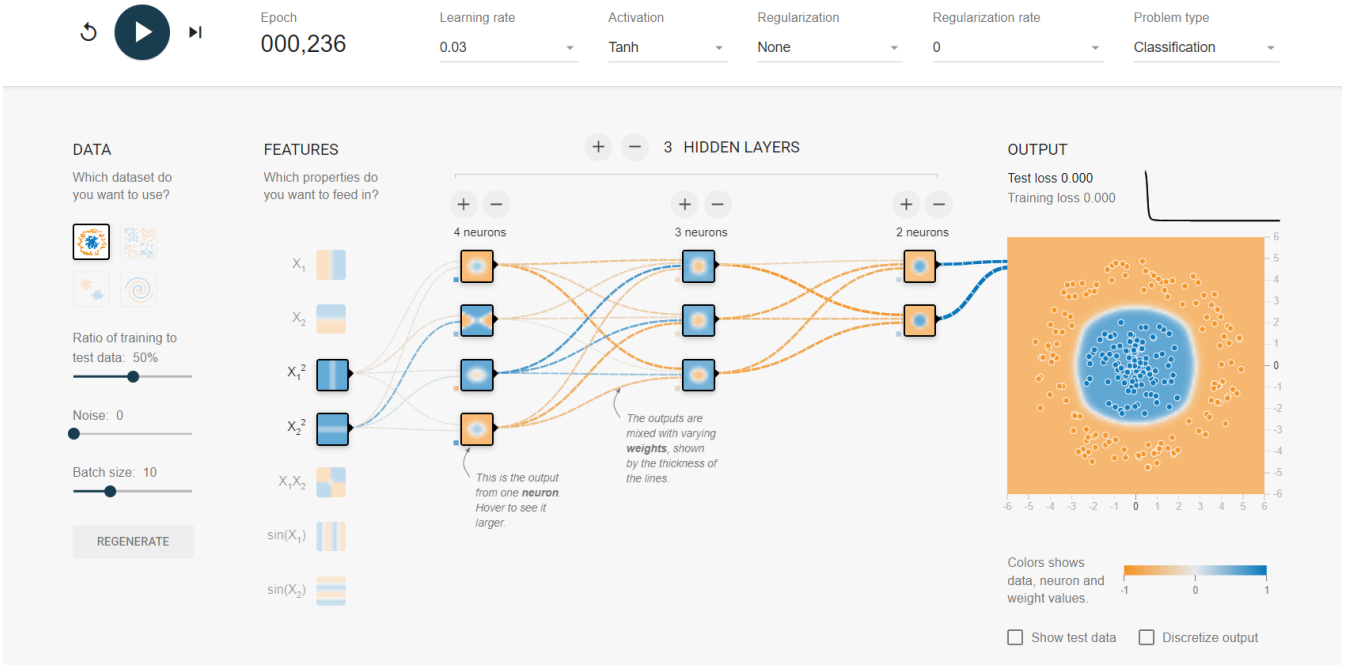
However, if decision boundary would have drawn as follows, minimum margin between training data and the decision boundary would be maximized where each distance is 1.



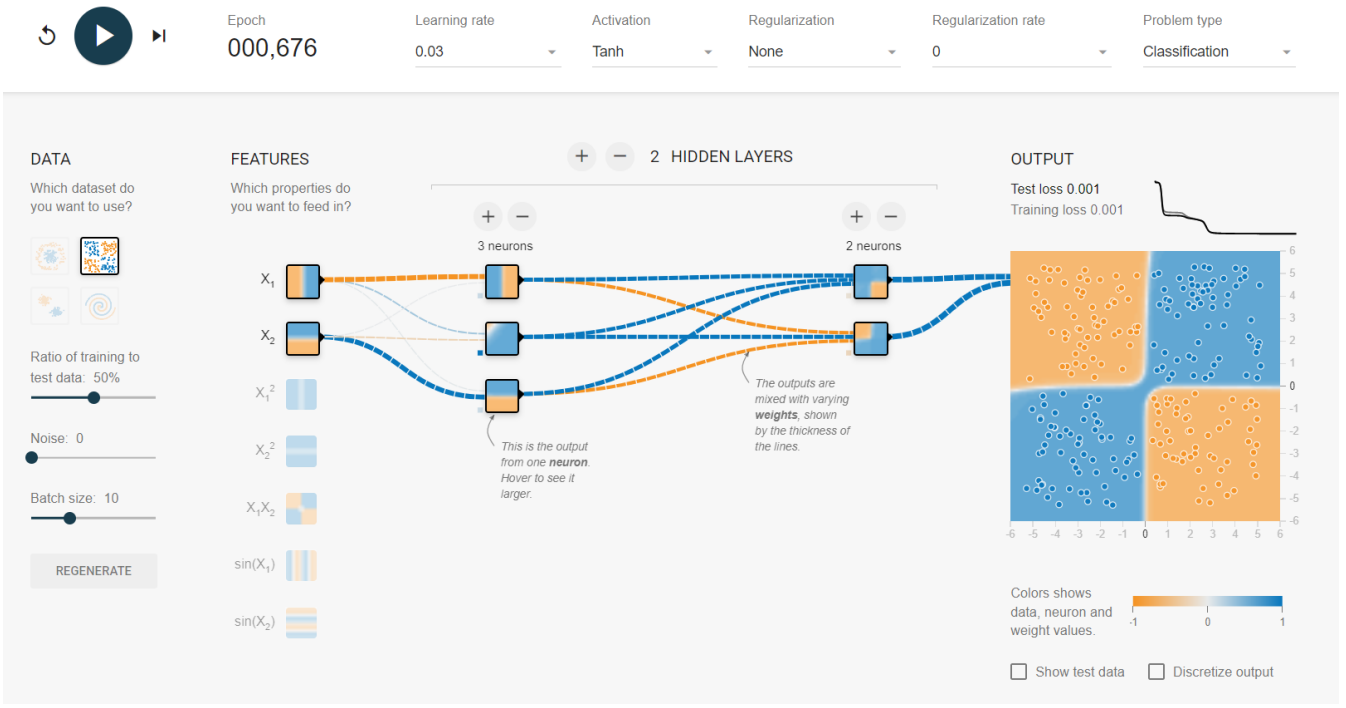
SOLUTION FOR QUESTION 2

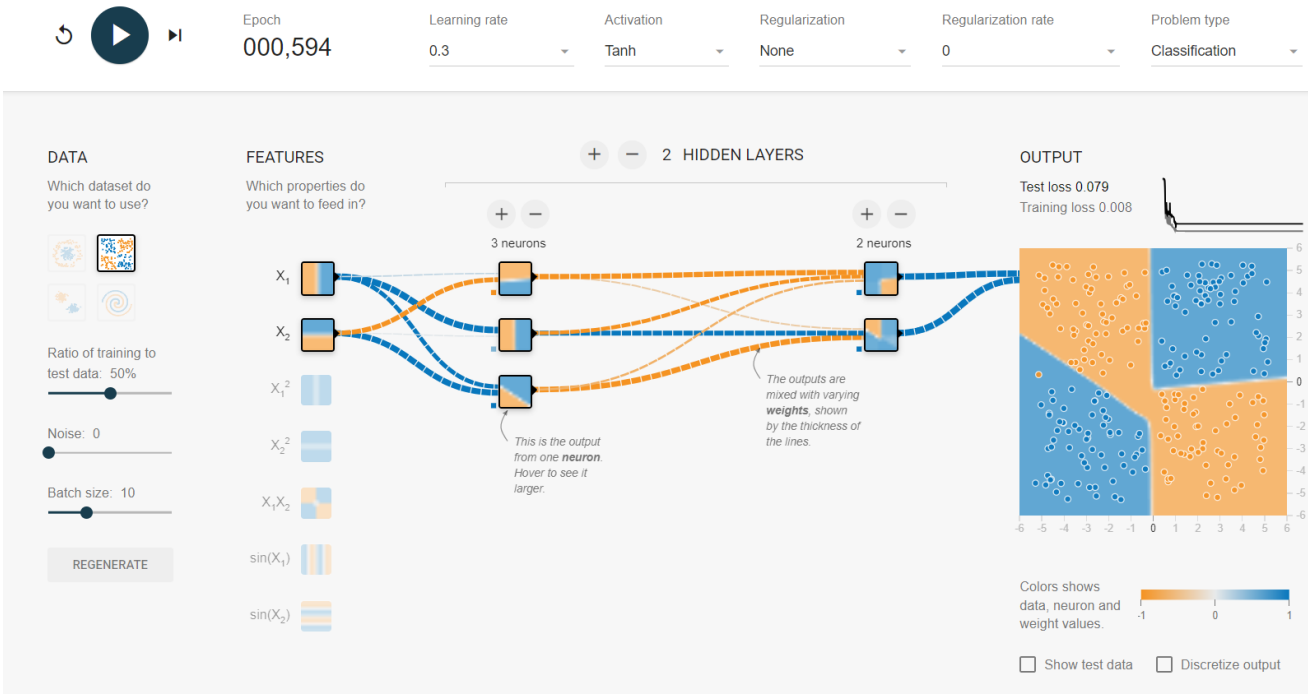
a)

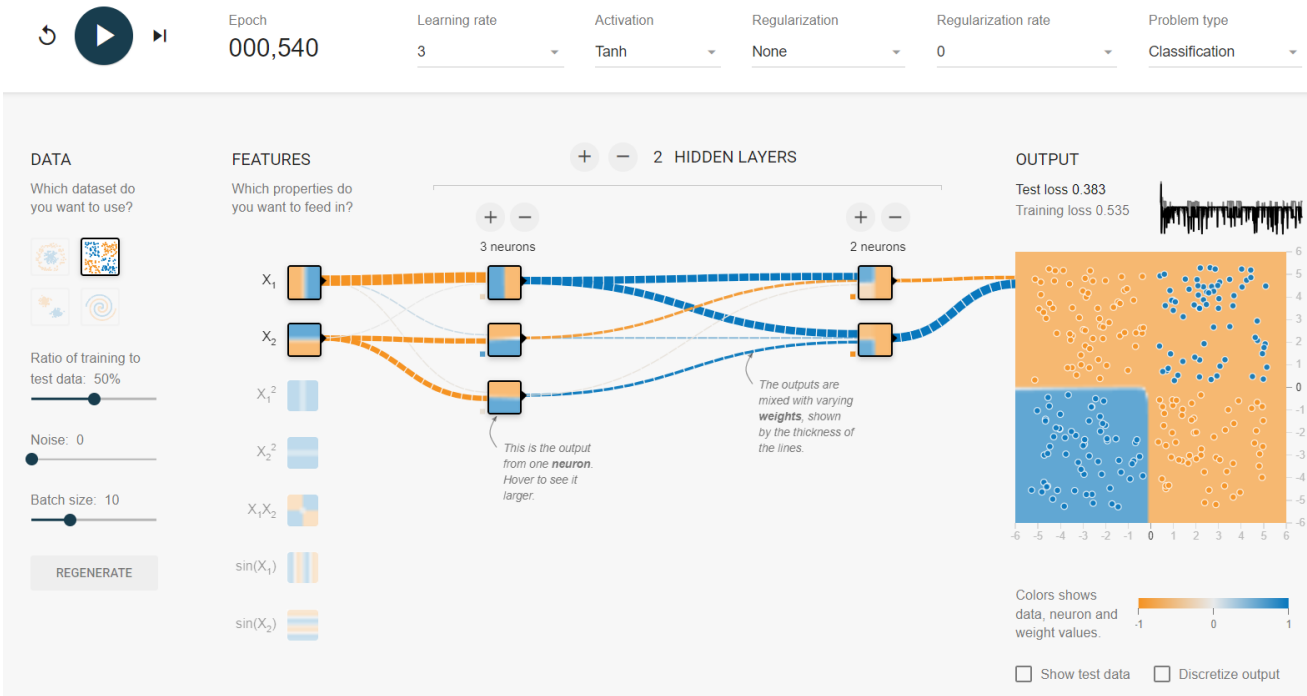



Epoch
000,236
Learning rate
0.03
Activation
Tanh
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
3 HIDDEN LAYERS
4 neurons
3 neurons
2 neurons
This is the output from one neuron. Hover to see it larger.
The outputs are mixed with varying weights, shown by the thickness of the lines.
OUTPUT
Test loss 0.000
Training loss 0.000
Colors shows data, neuron and weight values.
Show test data
Discretize output

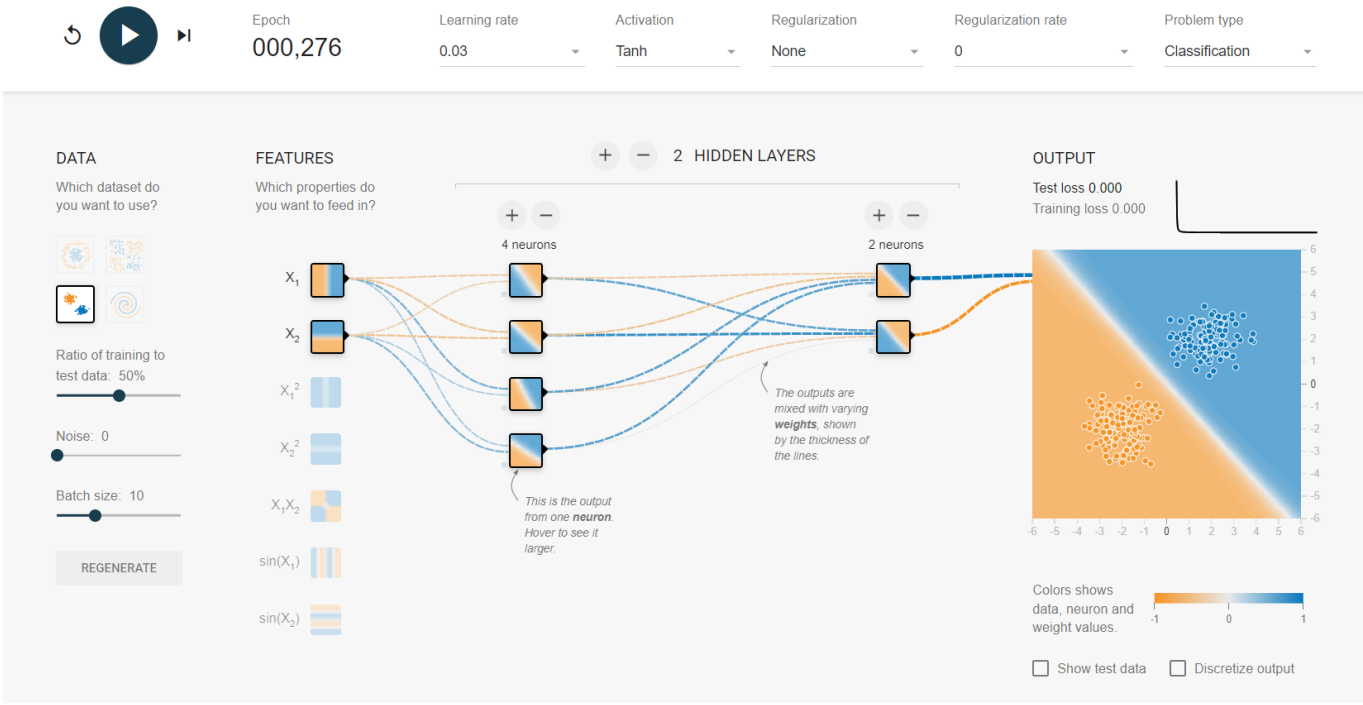
b)


Epoch
000,676
Learning rate
0.03
Activation
Tanh
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
2 HIDDEN LAYERS
3 neurons
2 neurons
This is the output from one neuron. Hover to see it larger.
The outputs are mixed with varying weights, shown by the thickness of the lines.
OUTPUT
Test loss 0.001
Training loss 0.001
Colors shows data, neuron and weight values.
Show test data
Discretize output

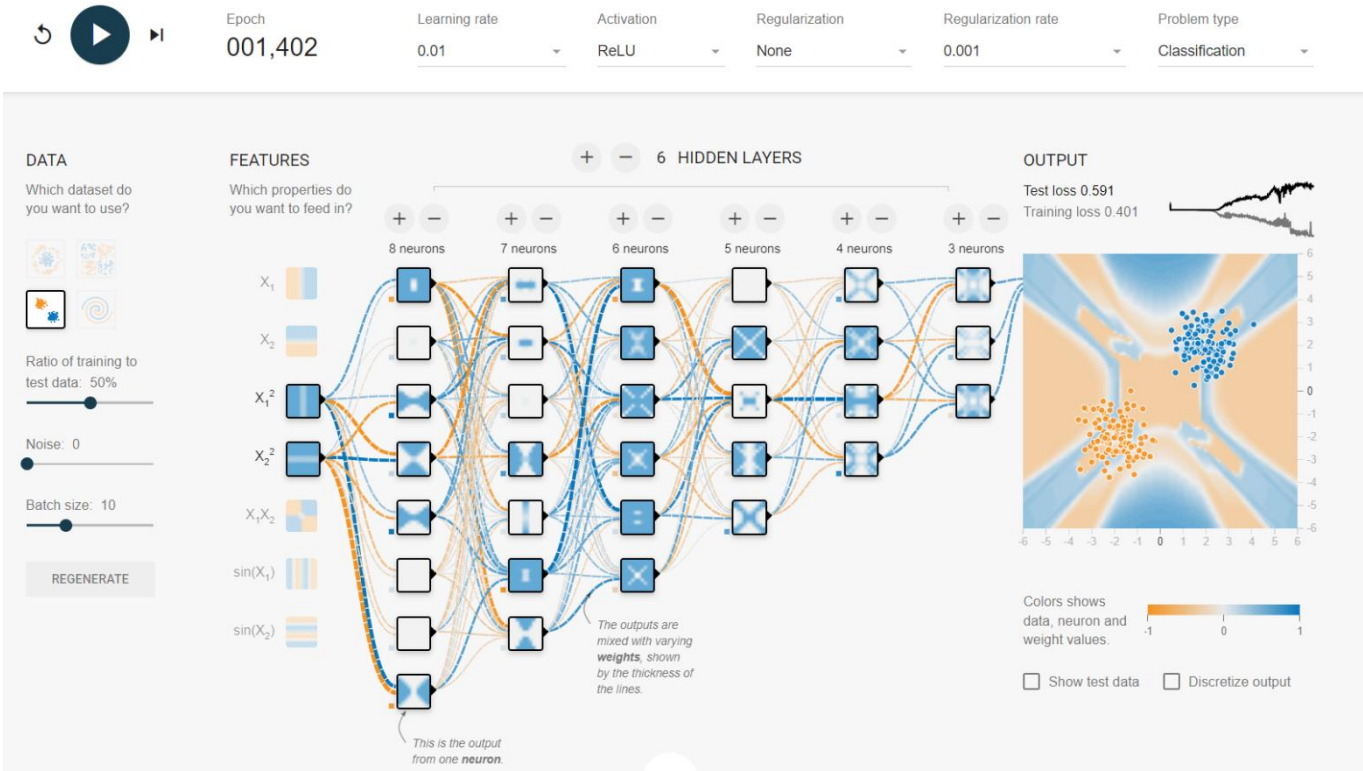

Epoch
000,594
Learning rate
0.3
Activation
Tanh
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
2 HIDDEN LAYERS
3 neurons
2 neurons
X1
X2
X1²
X2²
X1X2
sin(X1)
sin(X2)
This is the output from one neuron. Hover to see it larger.
The outputs are mixed with varying weights, shown by the thickness of the lines.
OUTPUT
Test loss 0.079
Training loss 0.008
Colors shows data, neuron and weight values.
Show test data
Discretize output


Epoch
000,540
Learning rate
3
Activation
Tanh
Regularization
None
Regularization rate
0
Problem type
Classification
DATA
Which dataset do you want to use?
Ratio of training to test data: 50%
Noise: 0
Batch size: 10
REGENERATE
FEATURES
Which properties do you want to feed in?
2 HIDDEN LAYERS
3 neurons
2 neurons
X1
X2
X1²
X2²
X1X2
sin(X1)
sin(X2)
This is the output from one neuron. Hover to see it larger.
The outputs are mixed with varying weights, shown by the thickness of the lines.
OUTPUT
Test loss 0.383
Training loss 0.535
Colors shows data, neuron and weight values.
Show test data
Discretize output

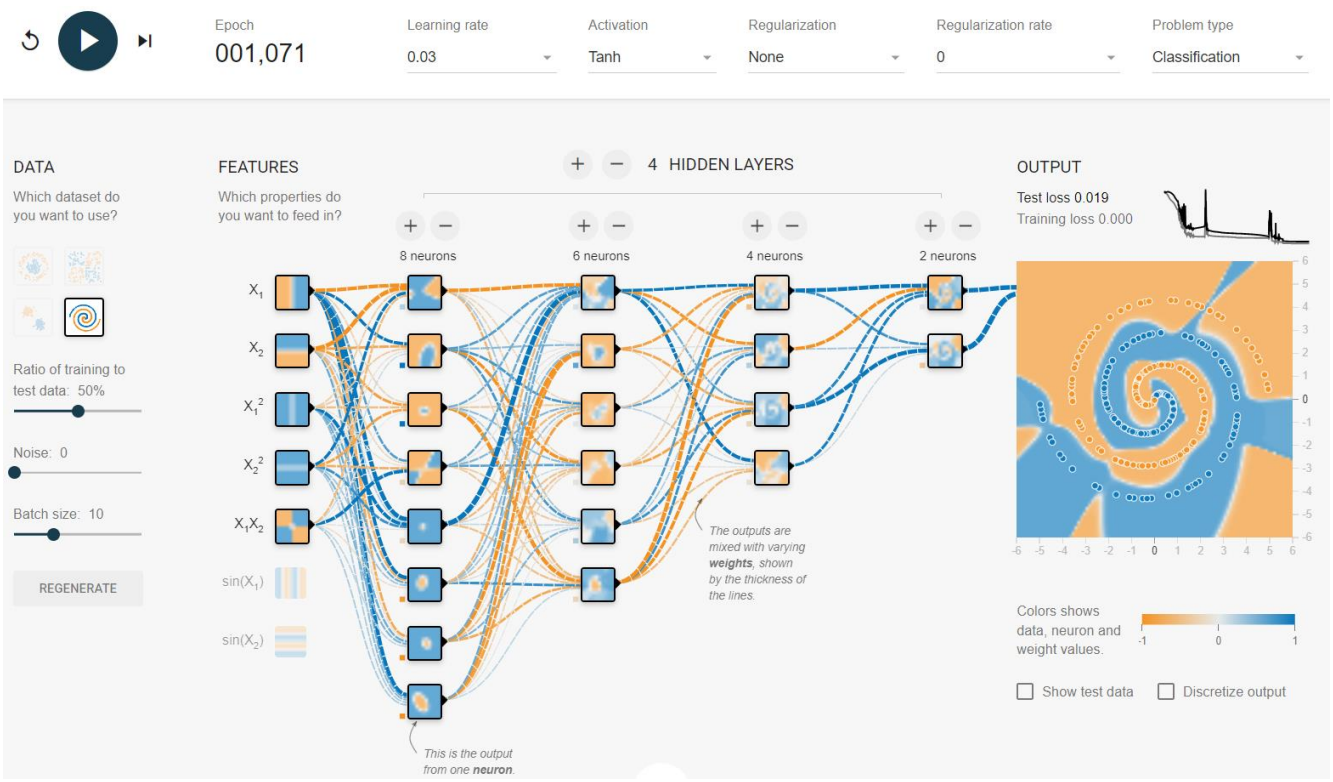
c)



Can not obtain when features are changed to X_1^2, X_2^2



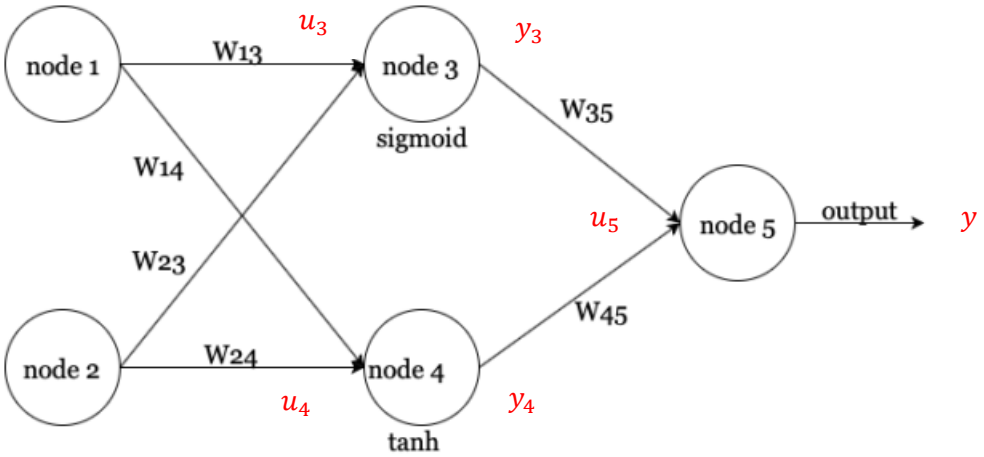
d)



SOLUTION FOR QUESTION 3

a)

The notations used during calculations are shown in following figure.



Then forward propagation can be calculated as follows

$$u_3 = (1)w_{13} + (-1)w_{23} = (1)(0.7) + (-1)(1.2) = -0.5$$

$$y_3 = \sigma(u_3) = \sigma(-0.5) = 0.37$$

$$u_4 = (1)w_{14} + (-1)w_{24} = (1)(-0.9) + (-1)(-1.1) = 0.2$$

$$y_4 = \tanh(u_4) = \tanh(0.2) = 0.19$$

$$u_5 = y_3 w_{35} + y_4 w_{45} = (0.37)(0.5) + (0.19)(-0.3) = 0.128$$

$$y = u_5 = 0.128$$

b)

Before calculating backpropagation, the error at the output layer can be calculated with following loss function

$$L(y, \hat{y}) = (\hat{y} - y)^2$$

In order to calculate backpropagation, gradient of Error with respect to each weight must be calculated. Gradients of each weight can be calculated as follows

$$\frac{\partial L}{\partial w_{45}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial u_5}\frac{\partial u_5}{\partial w_{45}} = 2(y - \hat{y})(1)y_4 = 2(0.128 - 1)(0.19) = -0.33$$

$$\frac{\partial L}{\partial w_{35}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial u_5}\frac{\partial u_5}{\partial w_{35}} = 2(y - \hat{y})(1)y_3 = 2(0.128 - 1)(0.37) = -0.65$$

$$\frac{\partial L}{\partial w_{13}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial u_5}\frac{\partial u_5}{\partial y_3}\frac{\partial y_3}{\partial u_3}\frac{\partial u_3}{\partial w_{13}} = 2(y - \hat{y})(1)w_{35}\sigma'(u_3)(1) = 2(0.128 - 1)(0.5)\big(1 - \sigma(-0.5)\big) = -0.55$$

$$\frac{\partial L}{\partial w_{14}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial u_5}\frac{\partial u_5}{\partial y_4}\frac{\partial y_4}{\partial u_4}\frac{\partial u_4}{\partial w_{14}} = 2(y - \hat{y})(1)w_{45}\tanh'(u_4)(1) = 2(0.128 - 1)(-0.3)(1 - \tanh^2(0.2)) = 0.50$$

$$\frac{\partial L}{\partial w_{23}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial u_5}\frac{\partial u_5}{\partial y_3}\frac{\partial y_3}{\partial u_3}\frac{\partial u_3}{\partial w_{23}} = 2(y - \hat{y})(1)w_{35}\sigma'(u_3)(-1) = -2(0.128 - 1)(0.5)\big(1 - \sigma(-0.5)\big) = 0.55$$

$$\frac{\partial L}{\partial w_{24}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial u_5}\frac{\partial u_5}{\partial y_4}\frac{\partial y_4}{\partial u_4}\frac{\partial u_4}{\partial w_{24}} = 2(y - \hat{y})(1)w_{45}\tanh'(u_4)(-1) = -2(0.128 - 1)(-0.3)(1 - \tanh^2(0.2)) = -0.50$$

Then, the weights can be updated as follows

$$w \leftarrow w - \eta\Delta$$

$$w_{45} = w_{45} - \eta\frac{\partial L}{\partial w_{45}} = (-0.3) - (0.1)(-0.33) \approx -0.27$$

$$w_{35} = w_{35} - \eta\frac{\partial L}{\partial w_{35}} = (0.5) - (0.1)(-0.65) \approx 0.57$$

$$w_{13} = w_{13} - \eta\frac{\partial L}{\partial w_{13}} = (0.7) - (0.1)(-0.55) \approx 0.76$$

$$w_{23} = w_{23} - \eta\frac{\partial L}{\partial w_{23}} = (1.2) - (0.1)(0.55) \approx 1.15$$

$$w_{24} = w_{24} - \eta\frac{\partial L}{\partial w_{24}} = (-1.1) - (0.1)(-0.55) \approx 1.05$$

w_{45}	w_{35}	w_{13}	w_{23}	w_{24}
-0.27	0.57	0.76	1.15	1.05

SOLUTION FOR QUESTION 4

DATASET CAN NOT FOUND